Data Extraction in ETL

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Q1: Define Data Transformation in ETL and explain why it is important.

Answer:

Data Transformation is the second stage of the ETL (Extract, Transform, Load) process. It involves converting raw data into a clean, consistent, and meaningful format before loading it into the target system or data warehouse.

Importance of Data Transformation

- Improves data quality by correcting errors and inconsistencies.
- Converts data into a standard format.
- Removes duplicate records.
- Combines data from multiple sources.
- Makes data suitable for reporting, analysis, and machine learning.

- Ensures consistency across different systems.

Example

Customer data collected from different branches may contain entries such as:

- Male
- male
- M

During transformation, all these values are standardized to "Male".

WHY:

Data transformation improves the accuracy, consistency, and usability of data, making it suitable for business intelligence and decision-making.

Q2: List any four common activities involved in Data Cleaning.

Answer:

Data cleaning is the process of improving data quality by identifying and correcting errors.

Four Common Data Cleaning Activities

1. Handling Missing Values

Missing values can be removed or replaced using suitable techniques such as mean, median, or mode imputation.

2. Removing Duplicate Records

Duplicate entries are identified and deleted to avoid inaccurate analysis.

3. Correcting Inconsistent Data

Different formats of the same value are standardized.

Example:

- Male
- male
- M

→ Converted to Male

4. Handling Outliers

Extreme values are detected and treated using methods such as capping or removal.

Q3: What is the difference between Normalization and Standardization?

Answer:

Basis	Normalization	Standardization
Purpose	Scales data to a fixed range	Centers data around the mean
Range	Usually 0 to 1	No fixed range
Formula	(X − Min) / (Max − Min)	(X − Mean) / Standard Deviation
Sensitive to Outliers	Yes	Less sensitive than normalization
Best Used	Neural Networks, Distance-based algorithms	Linear Regression, Logistic Regression, SVM

Example

Original values:

10, 20, 30

After Normalization:

0.0, 0.5, 1.0

After Standardization:

-1.22, 0.00, 1.22

Q4: A dataset has missing values in the "Age" column. Suggest two techniques to handle this and explain when they should be used.

Answer:

Two common techniques for handling missing values in the Age column are:

1. Mean Imputation

Replace missing values with the average age of all records.

Use When:

- Data is normally distributed.
- There are only a few missing values.

Example

Ages:

20, 25, 30, ?, 35

Mean:

27.5

Missing value becomes 27.5.

2. Median Imputation

Replace missing values with the median age.

Use When:

- The dataset contains outliers.
- Data is skewed.

Example

Ages: 18, 20, 21, 22, 90, ?

Median: 21 SO, Missing value becomes 21.

Q5: Convert the following inconsistent "Gender" entries into a standardized format ("Male", "Female"):

["M", "male", "F", "Female", "MALE", "f"]

Answer:

Original Values
["M", "male", "F", "Female", "MALE", "f"]

Standardized Values
["Male", "Male", "Female", "Female", "Male", "Female"]

Mapping Used

Original	Standardized
M	Male
male	Male
MALE	Male
F	Female
f	Female
Female	Female

Q6: What is One-Hot Encoding? Give an example with the categories: "Red, Blue, Green".

Answer:

One-Hot Encoding is a technique used to convert categorical data into numerical format by creating a separate binary column for each category.

Each category is represented by:

- 1 → Category present
- 0 → Category absent

Example

Original Data

Color
Red
Blue
Green

After One-Hot Encoding

Color_Red	Color_Blue	Color_Green
1	0	0
0	1	0
0	0	1

Q7: Explain the difference between Data Integration and Data Mapping in ETL.

Answer:

Data Integration	Data Mapping
Combines data from multiple sources into one unified dataset.	Defines how data from the source corresponds to the target system.
Focuses on merging data.	Focuses on matching fields between systems.
Occurs during the transformation stage.	Guides the transformation process.
Example: Combining sales data from multiple branches.	Example: Mapping "Cust_ID" in the source to "CustomerID" in the target database.

Example

Suppose data comes from two systems:

System A

Cust_ID

System B

CustomerID

During Data Mapping, these two fields are linked.During Data Integration, the records from both systems are combined into one dataset.

So, Data Integration combines information from different sources, while Data Mapping defines how fields from those sources correspond to the destination.

Q8: Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Answer:

When outliers are present, Z-score Standardization is generally preferred because it preserves the relative distribution of the majority of the data better than Min-Max Scaling, which is highly affected by extreme values.

Z-score Standardization transforms data using the mean and standard deviation, while Min-Max Scaling rescales data to a fixed range (usually 0 to 1).

Why Z-score is Preferred When Outliers Exist

- Outliers greatly affect the minimum and maximum values used in Min-Max Scaling.
- This causes most normal data points to be compressed into a very small range.
- Z-score Standardization uses the mean and standard deviation, making it less influenced by extreme values.
- Many machine learning algorithms perform better when standardized data is used.

Example:

Dataset: 10, 12, 15, 18, 500

Using Min-Max Scaling, the value 500 stretches the range, causing the other values to become very close to 0.

Using Z-score Standardization, the normal values remain more distinguishable despite the presence of the outlier.